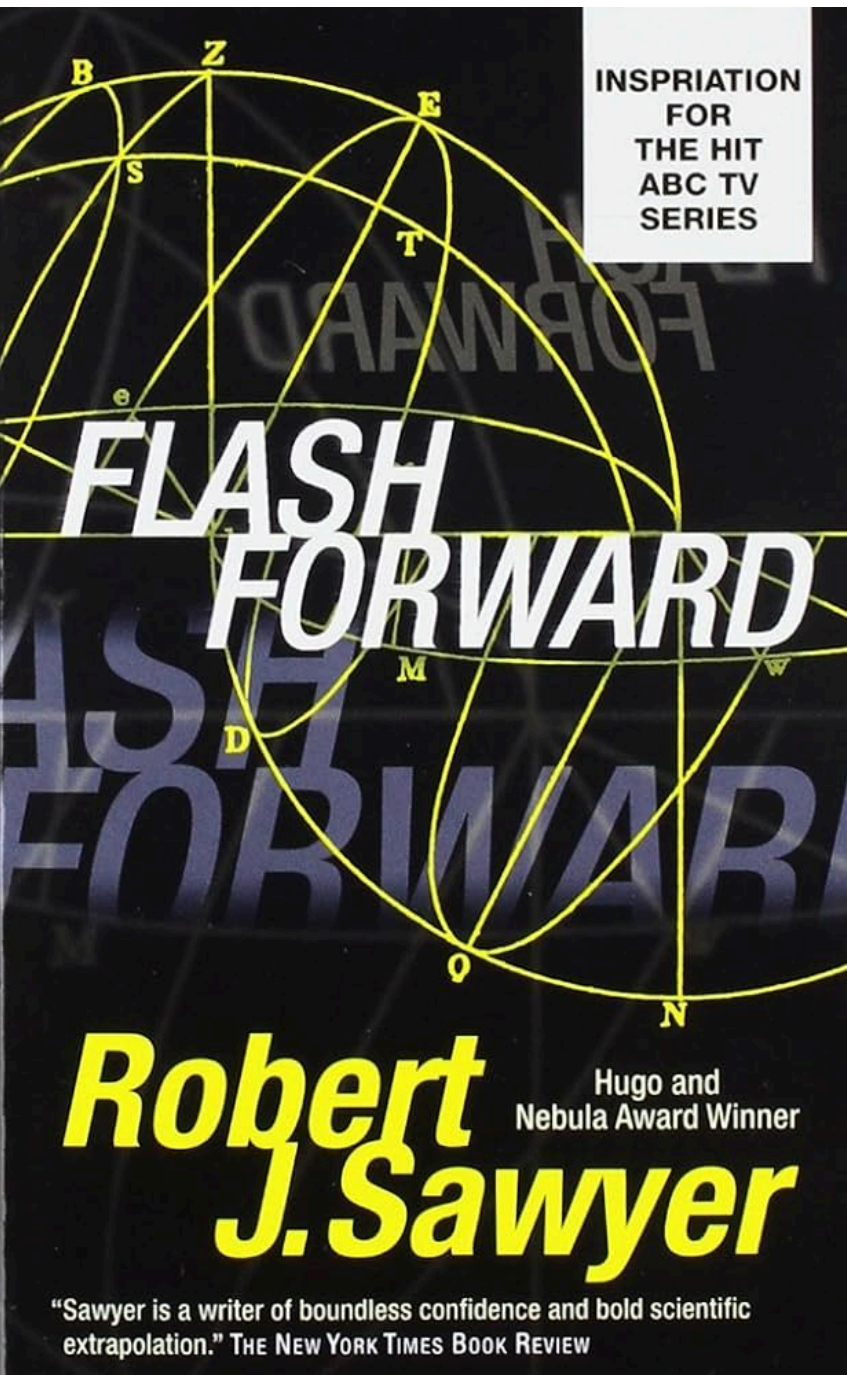




Indico: the 20 year history and evolution of an open-source project at CERN

Dominic Hollis & Tomas Roun - Indico Team (CERN)

What is CERN?



Largest particle physics lab in the world

- ~6.2 Square kilometers
- ~700 buildings on multiple sites
- 2500+ members of staff
- 12000+ visiting scientists
- 150'000+ visitors each year



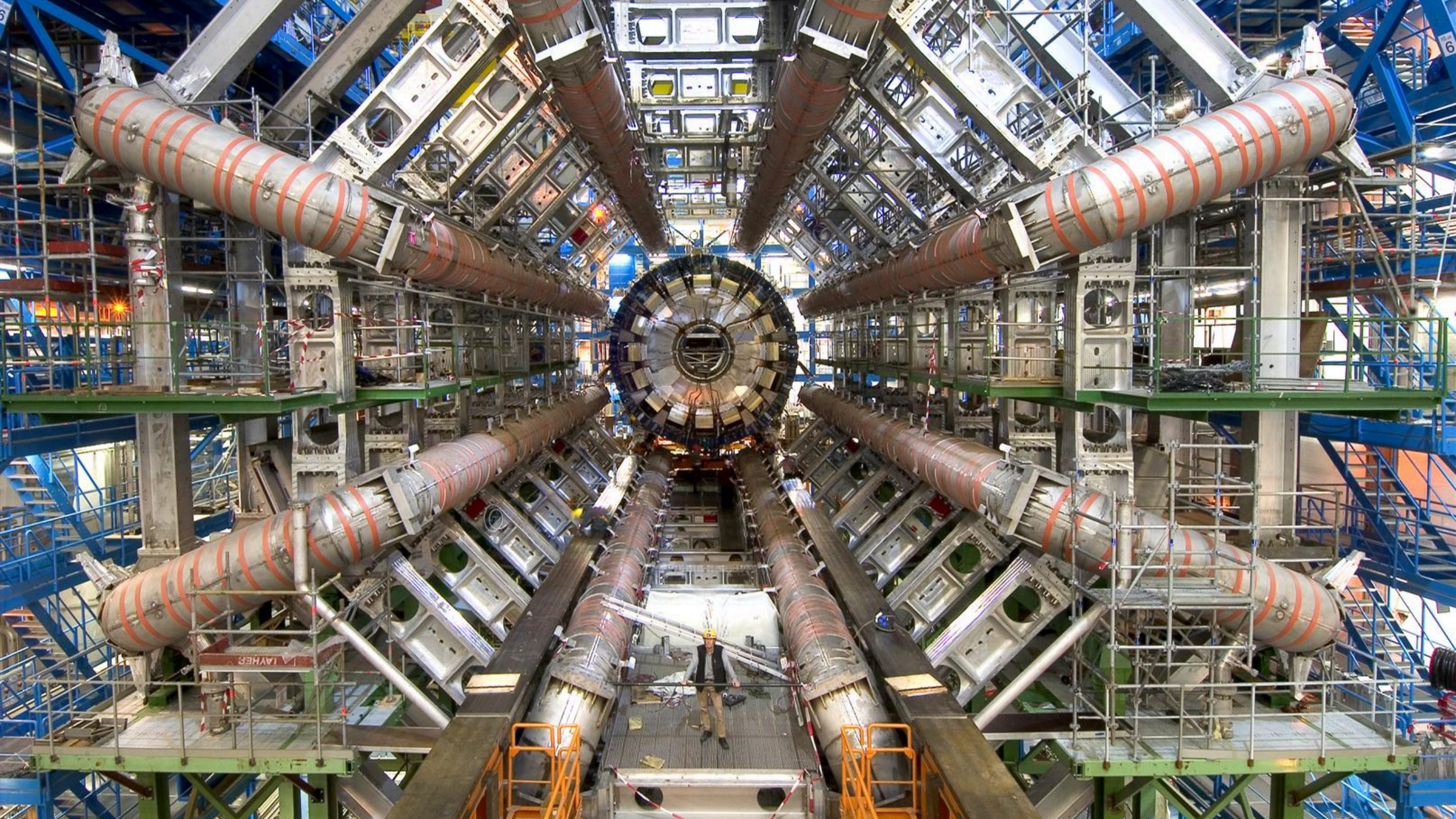
What exactly does CERN do?

CERN

PARTICLE

PARTICLE







TOM GAULD for NEW SCIENTIST

WHERE THE WEB WAS BORN

In the offices of this corridor, all the fundamental technologies of the World Wide Web were developed.

Started in 1990 from a proposal made by Tim Berners-Lee in 1989, the effort was first divided between an office in building 31 of the Computing and Networking Division (CN) and one in building 2 of the Electronics and Computing for Physics Division (ECP).

In 1991 the team came together in these offices, then belonging to ECP. It was composed of two CERN staff members, Tim Berners-Lee (GB) and Robert Cailliau (BE), aided by a number of Fellows, Technical Students, a Coopérant and Summer Students.

At the end of 1994 Tim Berners-Lee left CERN to direct the WWW Consortium (W3C), a world-wide organization devoted to leading the Web to its full potential. The W3C was founded with the help of CERN, the European Commission, the Massachusetts Institute of Technology (MIT), the Institut National pour la Recherche en Informatique et en Automatique (INRIA), and the Advanced Research Projects Agency (ARPA).

In 1995 Tim Berners-Lee and Robert Cailliau received the ACM Software System Award for the World Wide Web. In 2004, Tim Berners-Lee was awarded the first Millennium Technology Prize by the Finnish Technology Award Foundation.

The CERN Library
June 2004

A close-up portrait of Tim Berners-Lee, a middle-aged man with thinning grey hair, looking directly at the camera with a neutral expression. He is wearing a dark blue or black collared shirt. The background is a plain, light-colored wall.

Tim Berners-Lee
Web Developer

CERN ❤️ Open Source

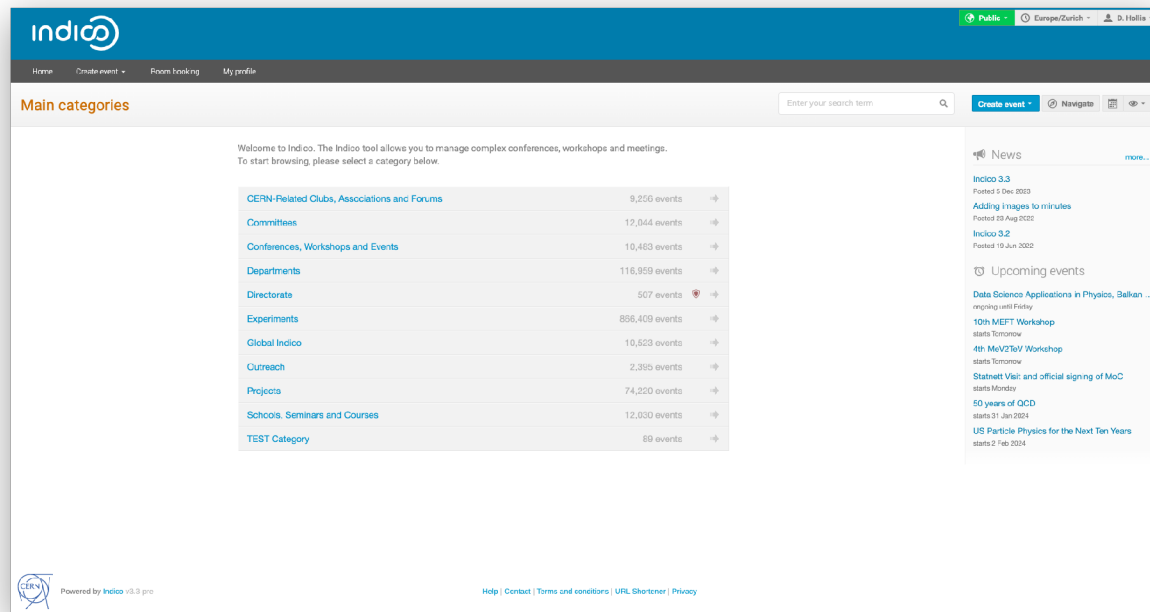


CERN ❤️ Open Source

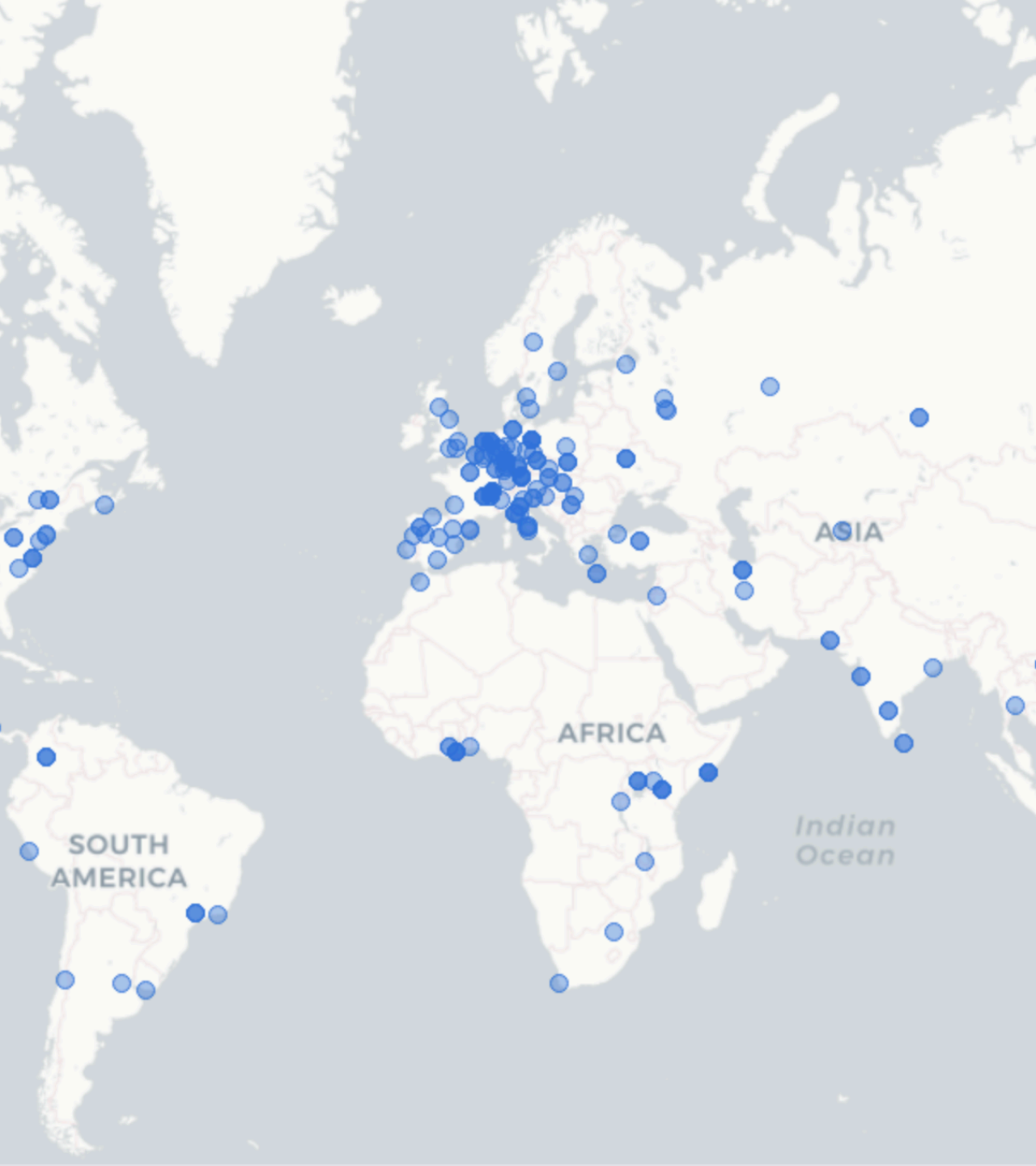
👉 <https://opensource.cern/>

👉 <https://github.com/CERN/awesome-cern>

What is Indico?



- **Event Management System**
- **Collaborative effort - MIT License**
- **Core Developed at CERN**
- **With contributions from the United Nations, Max-Planck Institute for Physics and many others!**
- **70+ developers** over the years



***The most popular event
management system you never
heard about***

- **300+ servers**
- **> 350K users**
- **20+ years** of development
- Initial growth in research, but growing beyond it
 - indico.un.org
 - events.canonical.org
 - indico.gnome.org
 - lpc.events

Under the hood

We ♥ Python

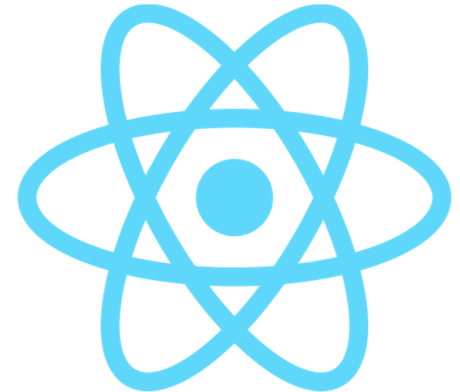
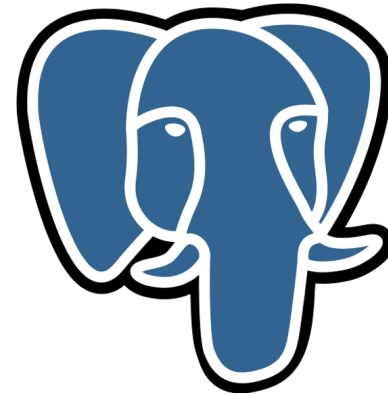


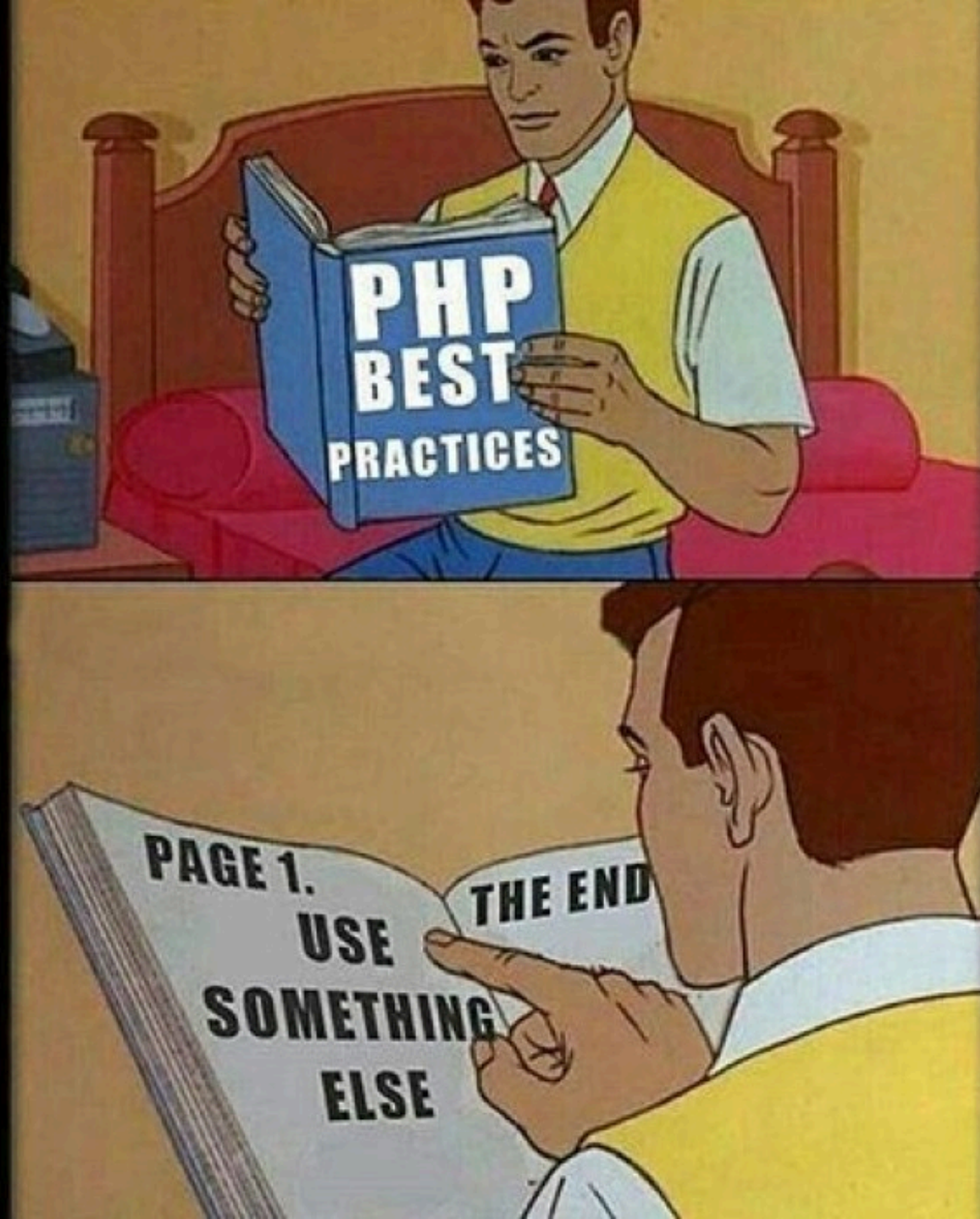
Flask

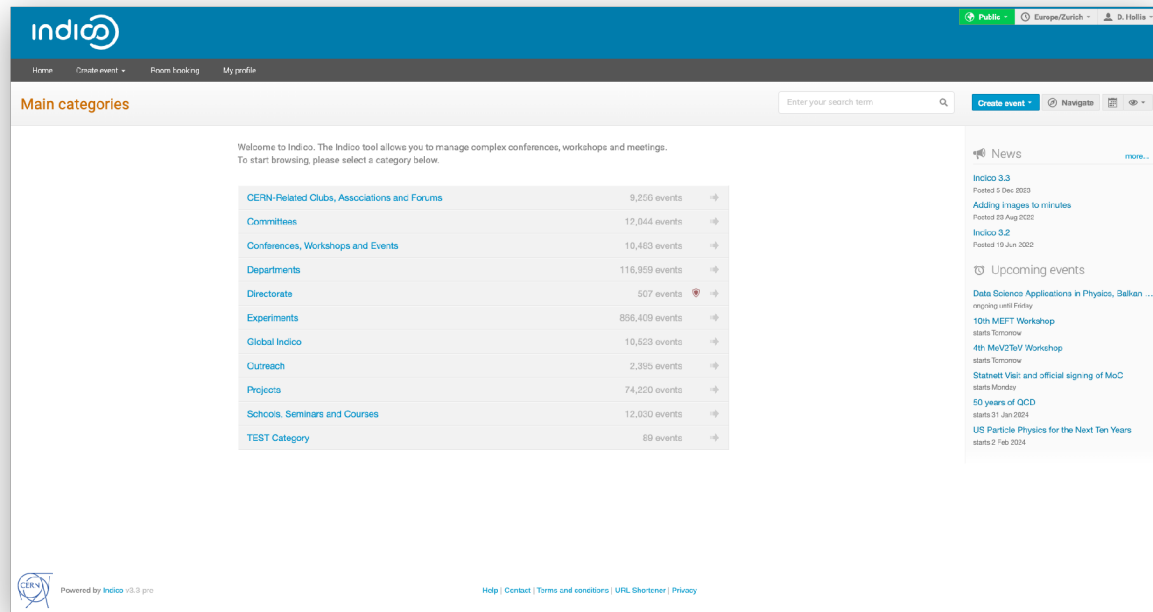
SQLAlchemy



python™







- **Event Management System**
- **Collaborative effort - MIT License**
- **Core Developed at CERN**
- **With contributions from the United Nations, Max-Planck Institute for Physics and many others!**
- **70+ developers** over the years



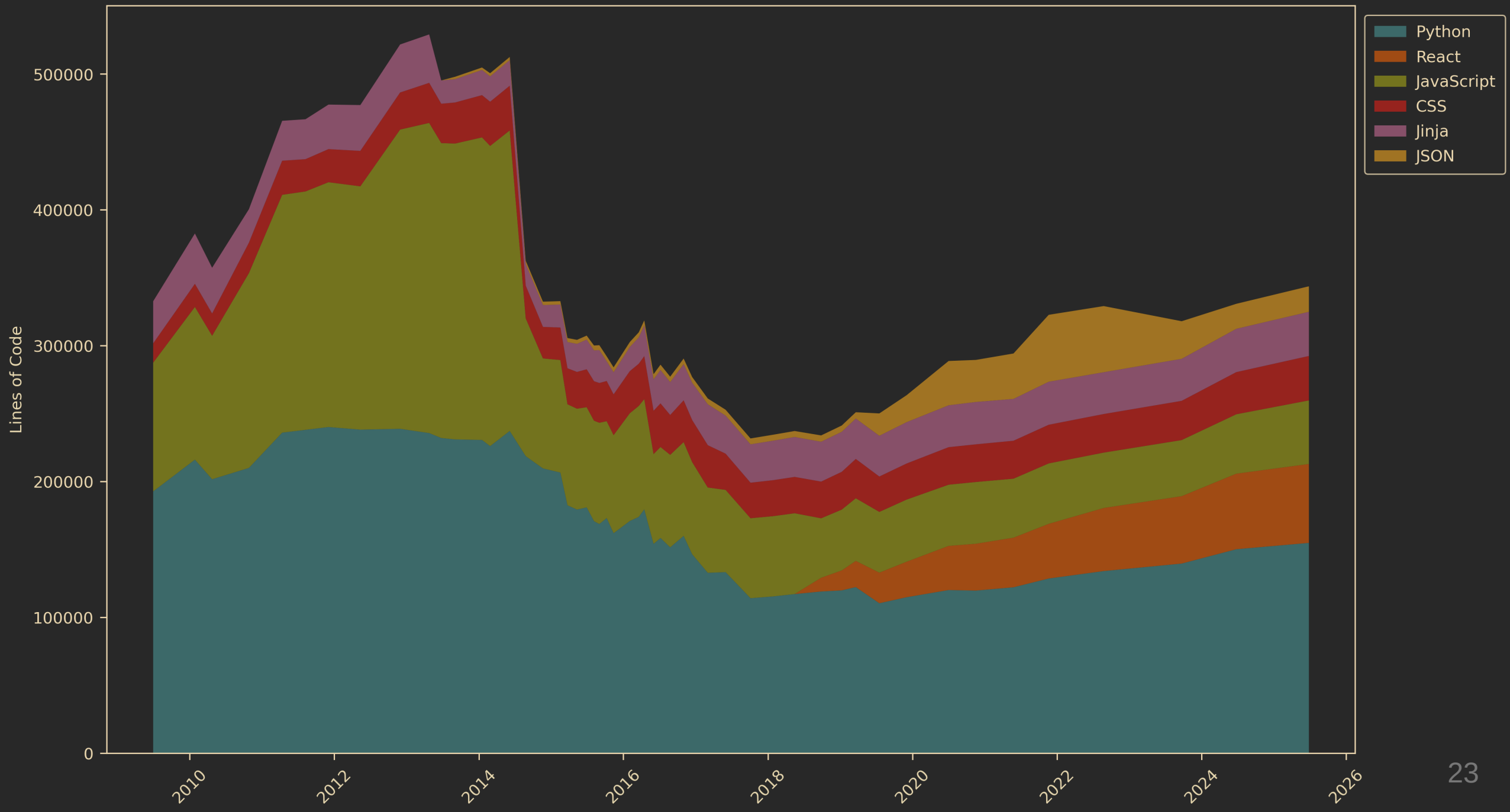
The most popular event management system you never heard about

- **300+ servers**
- **> 350K users**
- Initial growth in research, but growing beyond it
 - indico.un.org
 - events.canonical.org
 - indico.gnome.org
 - lpc.events

Stats

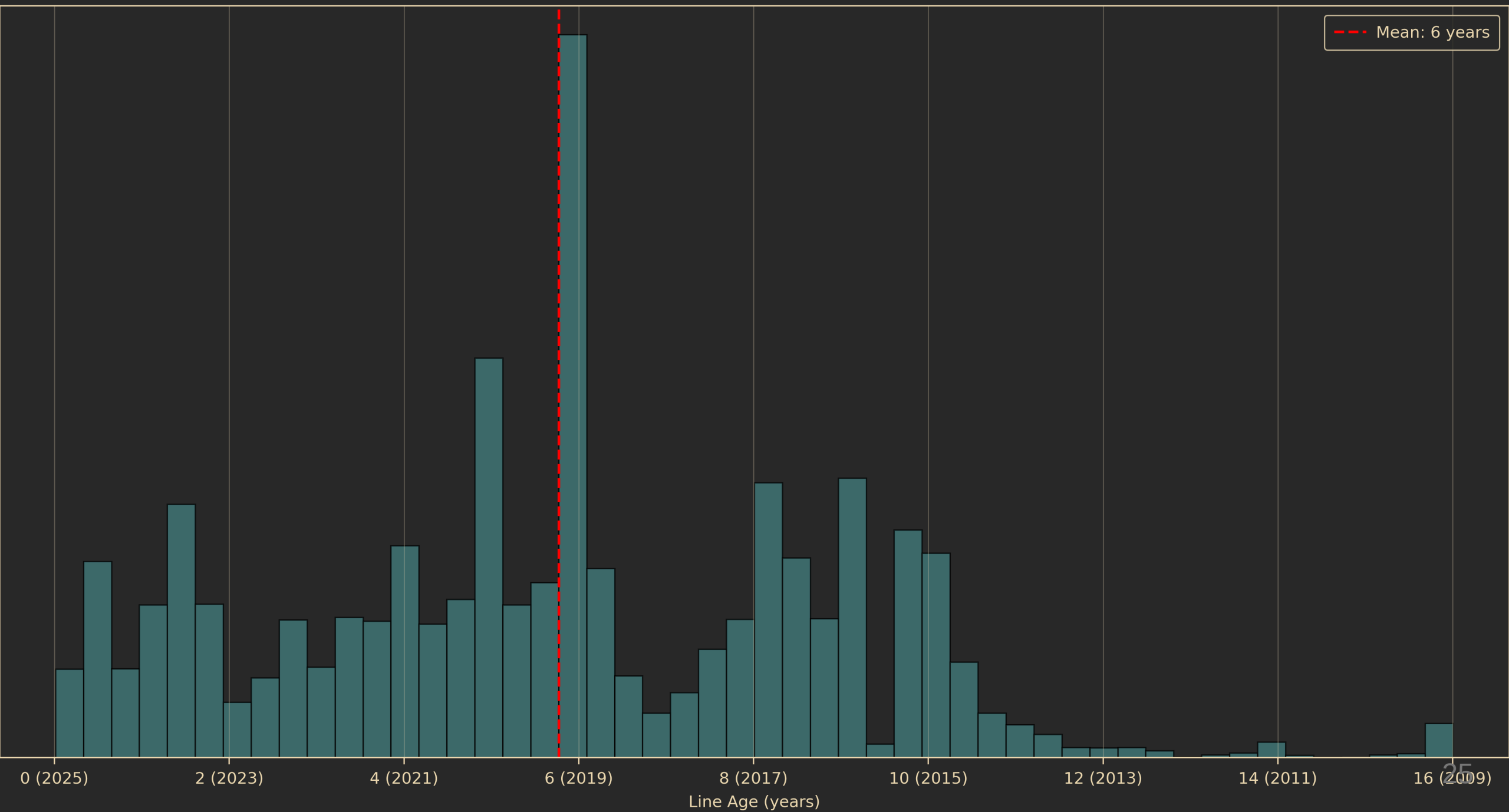
Code in Git since 2009. Migrated to GitHub in early 2015.

Language Composition Over Time



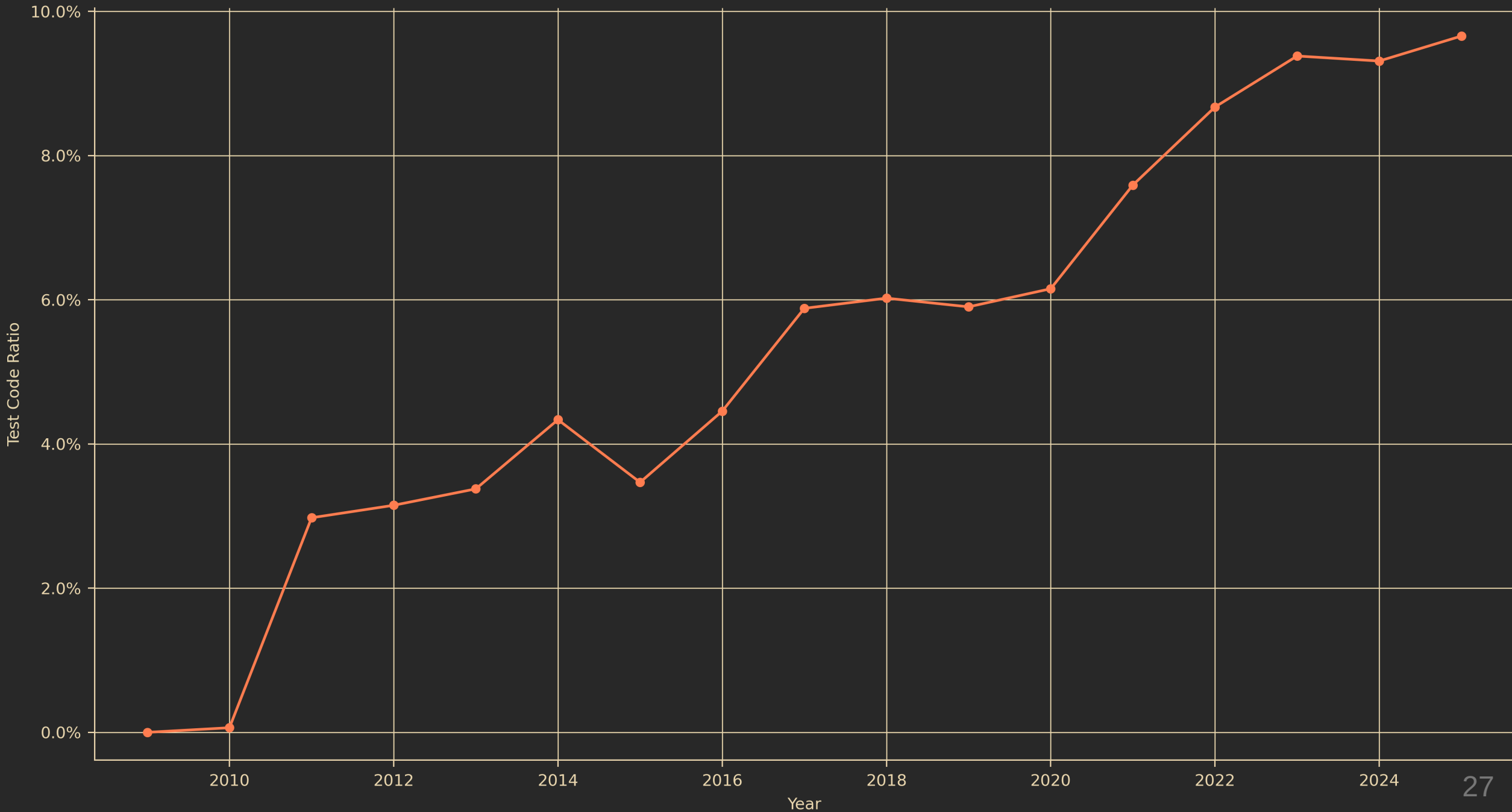
Dealing with Technical Debt

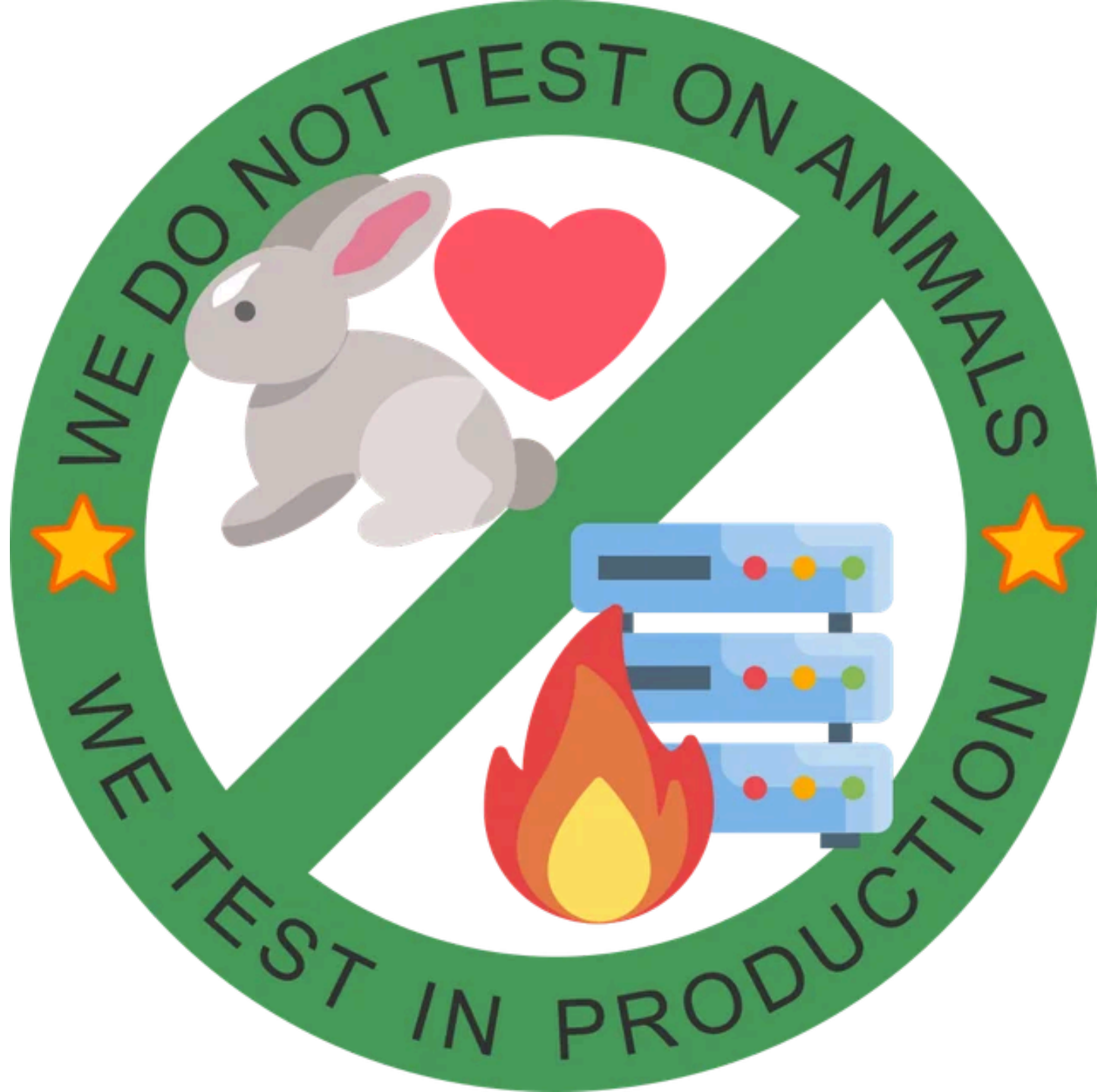
Line Age Distribution



Tests

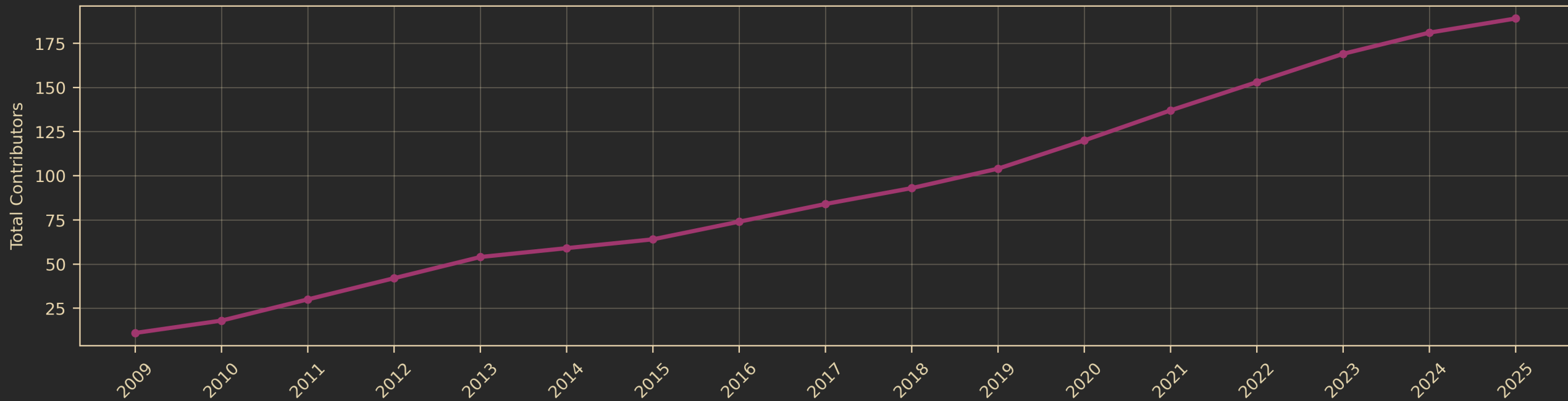
Test Code Ratio Over Time



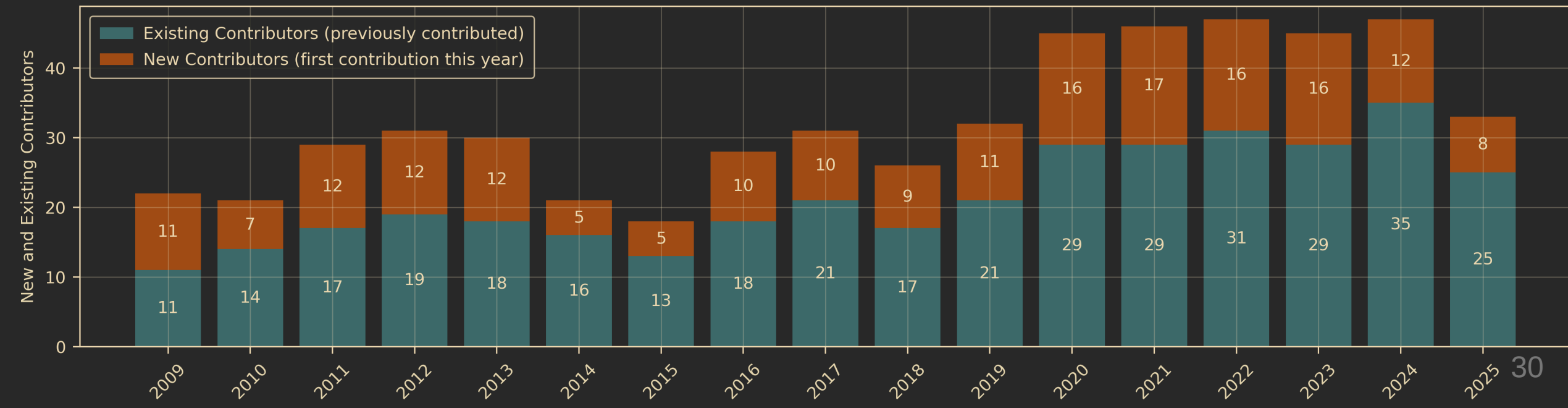


Contributors

All Contributors (Cumulative)



New Contributors

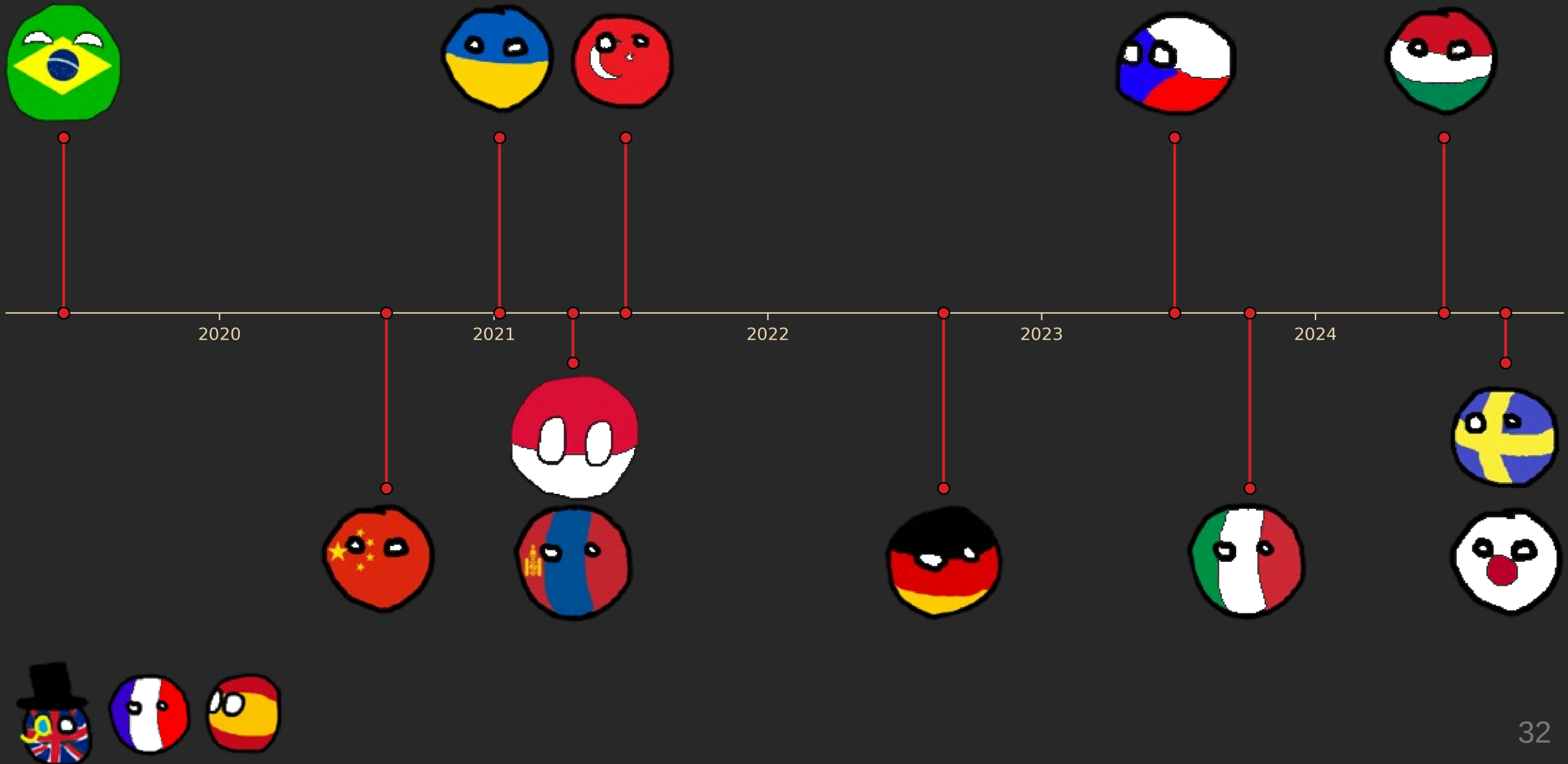


Translations

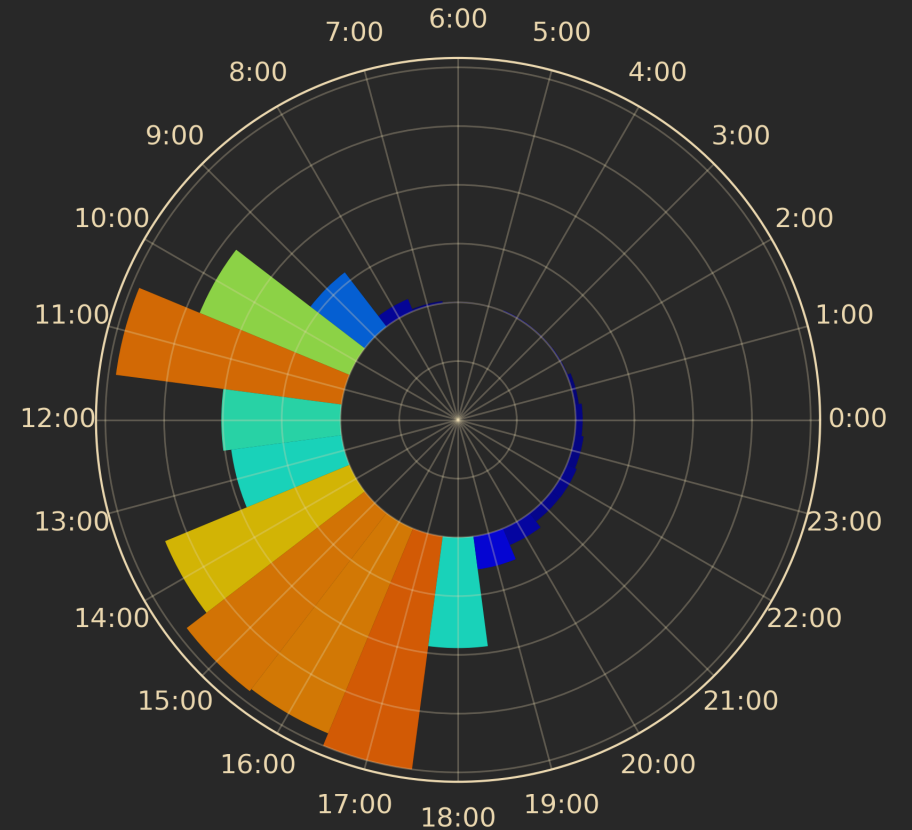
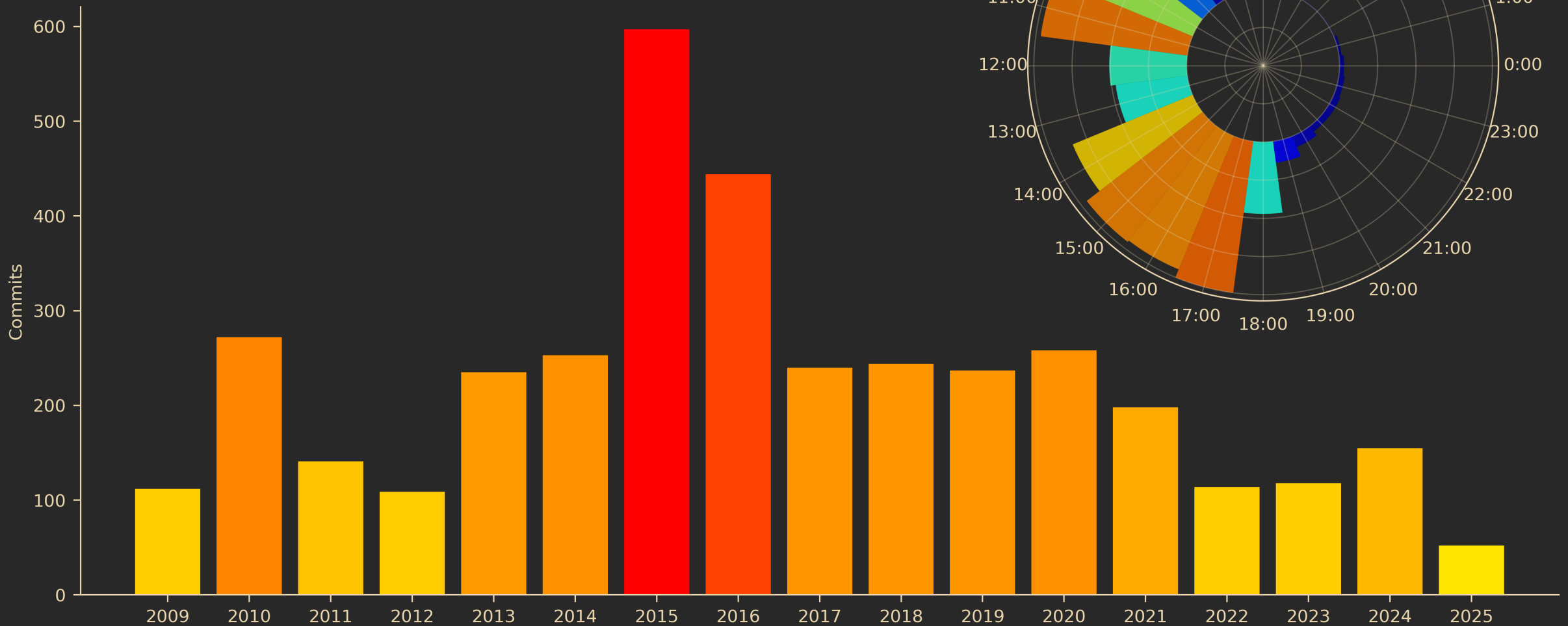
 Welcome to Indico
 Willkommen bei Indico
 Bienvenidos a Indico
 Bienvenue dans Indico
 Benvenuti in Indico
 Üdvözöljük az Indico-ban
 Witaj w Indico
 Bem vindo ao Indico

 Välkommen till Indico
 Indico'ya hoşgeldiniz
 Vítejte v Indicu
 Индикод тавтай морил
 Ласкаво просимо до Indico
 欢迎来到 Indico
 Indico 欢迎您的光临

Supported Languages Over Time



After-hours Commits





Lessons learned & Advice



How do you stay motivated after so many years?

How do you stay motivated after so many years?

By seeing the impact of your work. Seeing people use your project and appreciate the work you've done.

What advice would you give to other maintainers?

Keep the scope of the project in-mind. Do not blindly accept everything people ask for, especially if the maintenance burden is large.

What advice would you give to other maintainers?

Get some help - even if it is someone just looking at/filtering PRs, it can help a lot

 getindico.io

 [@getindico](https://matrix.org/#/indico:matrix.org)

 [@getindico](https://twitter.com/getindico)

 [#indico:matrix.org](https://matrix.org/#/indico:matrix.org)

indico